Route to Success
6th Annual Math Competition
Sample Questions

Note: Sample questions may not reflect overall difficulty of tests.

Division 1 (1st-2nd Grade) Sprint

1. There are 23 goldfish in a bowl. 9 of them jump out. How many are left?

   A. 13 B. 14 C. 12 D. 32

2. Each year, Joe gets the same number of birthday presents as his current age. When he is 68 years old, how many birthday presents will Joe have received over his entire life?

   A. 2056 B. 68 C. 2346 D. 1728

3. How much money is 2 quarters and 6 pennies?

   A. \$0.56 B. \$0.44 C. \$0.34 D. \$0.24

Division 1 Target

1. List the following numbers from least to greatest: 7, 0, -0.5, 12, -1.2

2. Jax has 50 marbles. One-fifth of them are red, 27 are green, and the rest are blue. How many red and blue marbles do they have in total?

3. Adam and Joan each have a pencil. Adam's pencil is 524 times longer than Joan's pencil. If Joan's pencil is 27 inches long, how long is Adam's pencil in feet? (1 foot = 12 inches).

Division 2 (3rd-5th Grade) Sprint

1. Find the mean of the following numbers: 4, 6, 7, 6, 4, 1, 9, 8, 3, 2, 5

   A. 55 B. 5 C. 44 D. 4

2. For how many positive integer values of n are both $\frac{n}{3}$ and $3n$ three-digit whole numbers?

   A. 12 B. 17 C. 21 D. 34

3. Suppose guavas are four times as expensive as peaches. Kaveh bought 3 guavas and 4 peaches for \$64. Beneatha paid for 7 guavas and 2 peaches with 2 hundred-dollar bills. How much change does Beneatha get back?

   A. \$40 B. \$80 C. \$120 D. \$160

Division 2 Target

1. If a pack of pencils costs \$4.50, and each pencil costs \$0.75, how many pencils are in the pack?
2. Add the fractions $\frac{3}{8}$ and $\frac{1}{3}$ and express the sum as a fraction in simplest form.
3. $ABCD$ is a square with area 64. Points E and F are midpoints of AB and BC respectively. What is the area of trapezoid $AEFC$?

Division 3 (6th-8th Grade) Sprint

1. Evaluate: $\frac{8^9}{4^{11} \cdot 2^3}$

   A. 4 B. $\frac{1}{8}$ C. $\frac{1}{4}$ D. 8

2. A circle is inscribed inside a square of side length 8. What is the area inside the square but outside the circle? Leave your answer in terms of π.

   A. 64 - 16π B. 64 - 4π C. 64 + 64π D. 64 + 4π

3. How many ways are there to rearrange the letters in "BOBCAT"?

   A. 360 B. 720 C. 180 D. 240

Division 3 Target

1. Arianna sheds 200 tears every single day. $\frac{2}{5}$ of those tears are due to Physics. $\frac{1}{2}$ are due to AMCs (math competitions), and $\frac{1}{10}$ are due to Chemistry. What is the difference between the number of tears shed for Physics and Chemistry?
2. 2 candles of varying heights are lit at the same time. After 5 minutes the longer one was 12 cm in height while the other one was 5 cm shorter than its original height. 7 minutes after the candles were lit, their heights were equal. The shorter candle burns half as fast as the longer one. What was the original length of the longer candle?
3. Ryle eats 1 chicken nugget, then x chicken nuggets, then x^2 chicken nuggets, then x^3 chicken nuggets, and so on forever. What does x have to be for Ryle to eat 6 chicken nuggets? Express your answer as a fraction in simplest form.